

Lots of question

Chapter 1%

Tutorial: 1% Me

- ② 1) Mention the relative advantage and disadvantage high level language and assembly language.
- ② 2) Advantages of Assembly language. slide. (12) 16c
- ② 3) Define microprocessor
- ② 4) Discuss the evolution of microprocessor
- ② 5) What are the differences between an accumulator based microprocessor and general purpose microprocessor.
- ② 6) Define RISC microprocessor
- ② 7) Differentiate between RISC and CISC type microprocessor.
- ② 8) Discuss PUSH and POP operation of microprocessor.
- ② 9) Discuss the 16 bit FLAG register of 16 bit microprocessor

core i7
25

10) What are the functions of CPU?

2 11) Mention the properties of INTEL 8086 microprocessor.

12) What are the differences between INTEL Core i5 and Core i7?

2 13) Draw the internal architecture of 8086 microprocessor.

2 14) Mention the purpose of BIU and EU.

② 15) Linear decoding. An 8-bit microprocessor interfaced to a 64K RAM system using the linear select decoding technique.

16) Draw the pin diagram of 8086 microprocessor (maximum mode or minimum mode).

2 17) Explain how physical address is calculated from logical address from 16-bit microprocessor.

② 18) Discuss the READ and WRITE cycle timing.

② 19) State and define the purpose of interrupt and draw interrupt vector.

20) What do you mean by addressing mode. Explain at least six address mode of 8086 microprocessor.

21) Draw block diagram of ~~803~~ 8237 DMA controller along with associated logic. Explain its role as slave and master mode.

22) Describe briefly synchronous and asynchronous communication.

23) Illustrate how 64 ~~bit~~ key, keyboard and 8 digit 7 segment display can be connected to a 8279.

24) Define 2 key lockout and N-key rollover. Draw the general architecture of the Intel 8279 keyboard and display controller.

25) Draw and describe internal architecture of 80286 mic. What are the program invisible register

* Protection mechanism of 80286

26) Write a short note on memory addressing modes of 80386.

27) What is difference between 80286 real address mode and protected virtual address mode?

28) How protected virtual address mode is used to calculate the physical address.

29) What is hyperthreading technology. Draw and describe internal architecture of Pentium microprocessor and also discuss features.

30) Give a truth table of Pin no 26, 27, 28. → Tut-2

31) Discuss main memory array design using

full/partial decoding technique. with advantage/disadvantage

32) What is the relationship between 8086 and 8284 input clock. Does 8086 have an on-chip clock circuitry?

Q.2/33) Construct a $(4K \times 8)$ RAM using $(4K \times 1)$ RAM chip. $(1K \times 1)$ RAM

Q.2/34) Why is it necessary at the start of an interrupt service procedure to PUSH all registers used in the procedure and to POP them at the end of the procedure.

Q.2/35) Explain how data can be transferred between main memory and an external I/O device using DMA controller.

~~Q.2/36)~~ Explain how the intel 80286 microprocessor translates logical address to physical address.

37) Explain the modes of operation determined by bits 7 and 6 of the 8237 channel's mode register.

Q.2/38) Math of addressing mode Tutorial 2

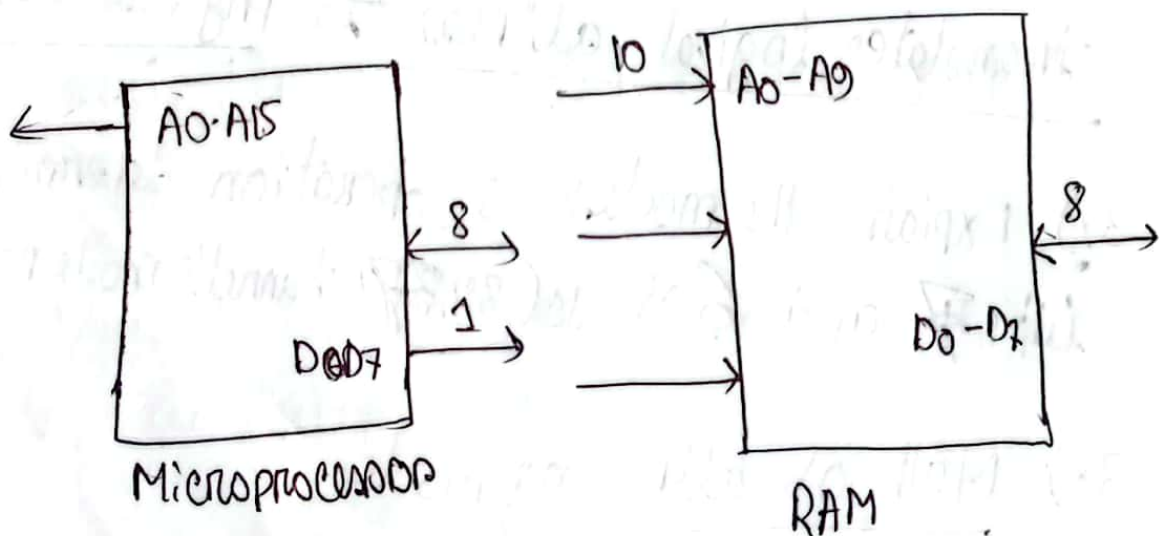
one large question for addressing mode. Tutorial 2

Tutorial 2

② 39) Write down the assembly language statements which will perform the following operations:

- i) Load the number 7986H into the BP register.
- ii) Copy the BP register contents to SP register.
- iii) Copy the contents of AX register to the

40) Assume the following microprocessor and the RAM chip. Draw a neat logic diagram showing connections between the microprocessor and RAM chip to obtain a memory of 6 KB and analyze the memory map. Use linear decoding technique.



2nd Tutorial

✓ 41 Explain the operation of the pin 35 to 38 of 8086 (in min mode)

42 Explain how ^{is an 8086} ~~interrupt vector table~~ enters into wait states, ~~what~~

43 What information is on the buses during wait states

44 Discuss the basic static RAM organization.

Tut 2

~~45~~ Explain how ~~data~~ can be transferred between main memory and external I/O using DMA controller

45 Explain the operation of 8284 A clock generator.

46 What is microcontroller. Internal Block diagram of microcontroller.

*47 Identifying addressing modes.

Tutorial 2 ① MOV DX, 92H

② MOV AX, DS:START

③ ADD DX, 9 [DI][BP]

48 Briefly discuss the instruction format of 8086 microprocessor.

49 Draw the basic DRAM organization and Discuss, and Also SRAM

Tutorial 2

50 Describe the interrupt response of an 8086 family processor.

51 Operation of an 8259A priority interrupt controller.

(Q251) Problem

52 Draw the description formats of 80286 and 80886 microprocessor

* Microprocessor Microcontroller and pipelining in RISC type microproc-

Description of General Purpose Registers